where $|\cdot|$ denotes the determinant and

$$C^{(1)}(\hat{\theta}) = \tfrac{\pi_{ab}}{2\pi(\hat{\theta})}\varsigma^{ab} - \big(\tfrac{h_{abcd}}{24} + \tfrac{h_{abc}\pi_u}{6\pi(\hat{\theta})}\big)\varsigma^{abcd} + \tfrac{h_{abc}h_{uef}}{72}\varsigma^{abcdef}, \tag{32}$$

$$\begin{aligned} C^{(2)}(\hat{\theta}) = {} & \tfrac{\pi_{abcd}}{24\pi(\hat{\theta})}\varsigma^{abcd} - \tfrac{\pi(\hat{\theta})h_{abcdef}+6h_{abcde}\pi_f+15h_{abcd}\pi_{ef}+20h_{abc}\pi_{def}}{720\pi(\hat{\theta})}\varsigma^{abcdef} \\ & + \tfrac{5\pi(\hat{\theta})h_{abcd}h_{efgh}+8\pi(\hat{\theta})h_{abcde}h_{fgh}+40h_{abc}\left(h_{defg}\pi_h+h_{def}\pi_{gh}\right)}{5760\pi(\hat{\theta})}\varsigma^{abcdefgh} \\ & - \tfrac{3\pi(\hat{\theta})h_{abcd}h_{efg}h_{hij}+4h_{abc}h_{def}h_{ghi}\pi_j}{5184\pi(\hat{\theta})}\varsigma^{abcdefghij} + \tfrac{h_{abc}h_{def}h_{ghi}h_{jkl}}{31104}\varsigma^{abcdefghijkl} \end{aligned} \tag{33}$$

where ς^{ab}, ς^{abcd}, ς^{abcdef}, $\varsigma^{abcdefgh}$, $\varsigma^{abcdefghij}$, and $\varsigma^{abcdefghijkl}$ represent the ab-th component of the second, the abcd-th component of the fourth, the abcdef-th component of the sixth moment, the abcdefgh-th component of the eigth moment, the abcdefghij-th component of the tenth moment, and the abcdefghijkl-th component of the twelfth moment, of the p dimensional random vector $Q \sim \mathcal{N}_p(0, \hat{I}(\hat{\theta})^{-1})$, respectively. ◇

Proof The proof is based on (i) Taylor-expanding the exponential of the log-likelihood of order five around $\hat{\theta}$, (ii) the definition of the exponential as a series and Taylor-expanding π to third order at the same point $\hat{\theta}$, and (iii) properties of the normal distribution.

Step (i) Let $h(\theta) = \frac{1}{n}\sum_{i=1}^{n} \log f(y_i \mid \theta)$, then since $h(\theta) \in C^6(\Theta)$ we know that there exists $\delta > 0$ such that in a ball $B_{\hat{\theta}}(\delta) \subset \mathbb{R}^p$ of radius δ centered at $\hat{\theta}$ the average log-likelihood $h_n(\theta)$ is well-approximated by a Taylor expansion of order 5. This combined with $\hat{\theta}$ being the MLE and the notation $\tilde{q} = \theta - \hat{\theta}$ yields

$$p(y^n) = \int_{\Theta} e^{-nh(\theta)}\pi(\theta)\mathrm{d}\theta = \int_{B_{\hat{\theta}}(\delta)} e^{-nh(\hat{\theta}) - \frac{nh_{ab}\tilde{q}^a\tilde{q}^b}{2} + \tilde{R}(\tilde{q})}\pi(\tilde{q})\mathrm{d}\tilde{q}, \tag{34}$$

$$= f(y^n \mid \hat{\theta}) \int_{B_{\hat{\theta}}(\delta)} e^{-\frac{nh_{ab}\tilde{q}^a\tilde{q}^b}{2!}} e^{-\tilde{R}(\tilde{q})}\pi(\tilde{q})\mathrm{d}\tilde{q}, \tag{35}$$

where

$$\tilde{R}(\tilde{q}) = n\big[\tfrac{h_{abc}\tilde{q}^a\tilde{q}^b\tilde{q}^c}{3!} + \tfrac{h_{abcd}\tilde{q}^a\tilde{q}^b\tilde{q}^c\tilde{q}^d}{4!} + \tfrac{h_{abcde}\tilde{q}^a\tilde{q}^b\tilde{q}^c\tilde{q}^d\tilde{q}^e}{5!} + \mathcal{O}(|\tilde{q}|^6)\big], \tag{36}$$

is the bounded remainder term since $h \in C^6(\Theta)$. The replacement of Θ by $B_{\hat{\theta}}(\delta)$ in the integral is justified if the mass is concentrated at $\hat{\theta}$, thus, whenever the integral with respect to the first order term falls off quadratically, that is, if

$$|n\hat{I}(\hat{\theta})|^{1/2} e^{-n(h(\theta)-h(\hat{\theta}))}\pi(\theta) = \mathcal{O}(n^{-2}), \tag{37}$$

which is the case when $\hat{\theta}$ is unimodal. When it is not unimodal, but $\hat{\theta}$ is a global maximum, then the condition implies that the requirement that the contribution of the other maxima is not too big.

Step (ii) After centering the integral at $\hat{\theta}$ we scale with respect to $\sqrt{n}$, that is, we apply the change of variable $q = \sqrt{n}\tilde{q}$, thus, $\int n^{-p/2}\mathrm{d}q = \int \mathrm{d}\tilde{q}$ and therefore

$$p(y^n) = (\tfrac{2\pi}{n})^{p/2} f(y^n \mid \hat{\theta}) |\hat{I}(\hat{\theta})|^{-1/2} \int_{B_{\hat{\theta}}(\sqrt{n}\delta)} \tilde{\varphi}(q) e^{-R(q)}\tilde{\pi}(q)\mathrm{d}q, \tag{38}$$

where $\tilde{\varphi}$ is the density of a multivariate normal distribution centered at 0 and covariance matrix $\Sigma = \hat{I}^{-1}(\hat{\theta})$, and where $\tilde{\pi}(q)$ is the Taylor approximation of π at the MLE, that is,

$$\tilde{\pi}(q) = \pi(\hat{\theta}) + \tfrac{\pi_a(\hat{\theta})q^a}{n^{1/2}} + \tfrac{\pi_{ab}(\hat{\theta})q^a q^b}{2!n} + \tfrac{\pi_{abc}(\hat{\theta})q^a q^b q^c}{3!n^{3/2}} + \mathcal{O}(n^{-2}), \tag{39}$$

and where the remainder term is now

$$R(q) = \frac{h_{abc}q^a q^b q^c}{3! n^{1/2}} + \frac{h_{abcd}q^a q^b q^c q^d}{4! n} + \frac{h_{abcde}q^a q^b q^c q^d q^e}{5! n^{3/2}} + \frac{h_{abcde}q^a q^b q^c q^d q^e q^f}{6! n^2} \mathcal{O}(n^{-5/2}).$$

To exploit the properties of Gaussian integrals we replace integration domain $B_{\hat{\theta}}(\sqrt{n}\delta)$ by $\mathbb{R}^p$, which is justified when n is large, and because the tails of a normal density fall off exponentially.

By definition of $e^{-R(q)}$ as a series and without the exponential approximation error

$$\begin{aligned} p(y^n) &\approx (\tfrac{2\pi}{n})^{p/2} f(y^n \mid \hat{\theta}) |\hat{I}(\hat{\theta})|^{-1/2} \\ &\times \int_{\mathbb{R}^p} \tilde{\varphi}(q) \big[1 - R(q) + \tfrac{R(q)^2}{2!} - \tfrac{R(q)^3}{3!} + \mathcal{O}(|R(q)|^4)\big] \tilde{\pi}(q) \mathrm{d}q. \end{aligned} \tag{40}$$

From here onwards we focus on the integral Eq. (40), which after some straightforward but tedious computations can be shown to be of the form

$$\int_{\mathbb{R}^p} \tilde{\varphi}(q) \big[A_0 + A_1 n^{-1/2} + A_2 n^{-1} + A_3 n^{-3/2} + A_4 n^{-2} + \mathcal{O}(n^{-3})\big] \mathrm{d}q, \tag{41}$$

where the A_j terms are functions of q and $\hat{\theta}$ defined by the series representation of $e^{-R(q)}$ and $\tilde{\pi}(q)$.

Step (iii) The terms A_j are given below. Of the following results only the exact values of A_0, A_2 and A_4 matter; what matters for A_1 and A_3 is that they only involve odd powers of q:

$$A_0 = \pi(\hat{\theta}) \tag{42}$$

$$A_1 = \pi_a q^a - \tfrac{h_{abc}\pi(\hat{\theta})}{6} q^a q^b q^c \tag{43}$$

$$A_2 = \tfrac{\pi_{ab}}{2} q^a q^b - \big(\tfrac{\pi(\hat{\theta}) h_{abcd}}{24} + \tfrac{h_{abc}\pi_u}{6}\big) q^a q^b q^c q^d + \tfrac{\pi(\hat{\theta}) h_{abc} h_{uef}}{72} q^a q^b q^c q^d q^e q^w \tag{44}$$

$$\begin{aligned} A_3 &= \tfrac{\pi_{abc}}{6} q^a q^b q^c - \tfrac{6h_{abcde}\pi(\hat{\theta}) + 30 h_{abcd}\pi_v + 60 h_{abc}\pi_{uv}}{720} q^a q^b q^c q^d q^e \\ &+ \tfrac{h_{abc}h_{uefl}\pi(\hat{\theta}) + 2h_{abc}h_{uef}\pi_l}{144} q^a q^b q^c q^d q^e q^w q^l \end{aligned} \tag{45}$$

$$\begin{aligned} A_4 &= \tfrac{\pi_{abcd}}{24} q^a q^b q^c q^d \\ &- \tfrac{\pi(\hat{\theta}) h_{abcdef} + 6h_{abcde}\pi_f + 15 h_{abcd}\pi_{ef} + 20 h_{abc}\pi_{def}}{720} q^a q^b q^c q^d q^e q^f \\ &+ \tfrac{5\pi(\hat{\theta}) h_{abcd} h_{efgh} + 8\pi(\hat{\theta}) h_{abcde} h_{fgh} + 40 h_{abc}\left(h_{defg}\pi_h + h_{def}\pi_{gh}\right)}{5760} q^a q^b q^c q^d q^e q^f q^g q^h \\ &- \tfrac{3\pi(\hat{\theta}) h_{abcd} h_{efg} h_{hij} + 4 h_{abc} h_{def} h_{ghi} \pi_j}{5184} q^a q^b q^c q^d q^e q^f q^g q^h q^i q^j \\ &+ \tfrac{\pi(\hat{\theta}) h_{abc} h_{def} h_{ghi} h_{jkl}}{31104} q^a q^b q^c q^d q^e q^f q^g q^h q^i q^j q^k q^l. \end{aligned} \tag{46}$$

Since for k odd A_k only involve odd powers of q we conclude that their integral with respect to $\tilde{\varphi}(q)$ vanishes. Hence,

$$p(y^n) = (\tfrac{2\pi}{n})^{p/2} f(y^n \mid \hat{\theta}) |\hat{I}(\hat{\theta})|^{-1/2} \pi(\hat{\theta}) \big[1 + \tfrac{E[A_2]}{n\pi(\hat{\theta})} + \tfrac{E[A_4]}{n^2 \pi(\hat{\theta})} \mathcal{O}(n^{-3})\big], \tag{47}$$

where $E[A_2]$ and $E[A_4]$ are expectations with respect to $Q \sim \mathcal{N}(0, \hat{I}(\hat{\theta})^{-1})$. This implies that the order n^{-1} and n^{-2} terms in the assertion are $C^{(1)}(\hat{\theta}) = E[A_2]/\pi(\hat{\theta})$ and $C^{(2)}(\hat{\theta}) = E[A_4]/\pi(\hat{\theta})$.

The components of higher moments can be expressed in terms of the covariances $\varsigma^{ab} = \mathrm{Cov}(Q^a, Q^b)$ using Isserlis' formula (Isserlis, 1918; McCullagh, 2018). For moments $\varsigma^{a_1 \cdots a_w}$, that is, a component of the wth moment of Q with $w = 2v$ even, the following holds

$$\varsigma^{a_1 \cdots a_w} = \sum_{u \in P_w^2} \prod_{i,j \in u} \varsigma^{ij}, \tag{48}$$

where P_w^2 is the collection of all pairs of which there are v. For instance, for $w = 4$, ς^{abcd} is a sum of 2-products of pairs, for $w = 6$ is a sum of 3-products of ς^{abcdef} and so forth and so on. More specifically,

$$\varsigma^{abcd} = \varsigma^{ab}\varsigma^{cd} + \varsigma^{ac}\varsigma^{bd} + \varsigma^{ad}\varsigma^{bc} \tag{49}$$

$$\begin{aligned}\varsigma^{abcdef} = & \varsigma^{ab}\varsigma^{cd}\varsigma^{ef} + \varsigma^{ab}\varsigma^{ce}\varsigma^{df} + \varsigma^{ab}\varsigma^{cf}\varsigma^{de} \\ & + \varsigma^{ac}\varsigma^{bd}\varsigma^{ef} + \varsigma^{ac}\varsigma^{be}\varsigma^{df} + \varsigma^{ac}\varsigma^{bf}\varsigma^{de} \\ & + \varsigma^{ad}\varsigma^{bc}\varsigma^{ef} + \varsigma^{ad}\varsigma^{be}\varsigma^{cf} + \varsigma^{ad}\varsigma^{bf}\varsigma^{ce} \\ & + \varsigma^{ae}\varsigma^{bc}\varsigma^{df} + \varsigma^{ae}\varsigma^{bd}\varsigma^{cf} + \varsigma^{ae}\varsigma^{bf}\varsigma^{cd} \\ & + \varsigma^{af}\varsigma^{bc}\varsigma^{de} + \varsigma^{af}\varsigma^{bd}\varsigma^{ce} + \varsigma^{af}\varsigma^{be}\varsigma^{cd},\end{aligned} \tag{50}$$

where all indexes $a, b, c, d, e, f = 1, 2, \ldots, p$. The expression of $\varsigma^{abcdefgh}$, $\varsigma^{abcdefghij}$, and $\varsigma^{abcdefghijkl}$ define sums of $105 = 3\times5\times7$, $945 = 3\times5\times7\times9$, and $10,395 = 3\times5\times7\times9\times11$ terms respectively, and due to space restrictions their exact forms are not displayed here.